

ABOLFAZL MOHAMMADIJOO

OBJECTIVE

Control and robotics engineer, software developer, researcher, and technical educator with a Mechanical Engineering background and more than a decade of independent project experience. I develop engineering software, control algorithms, numerical simulations, research prototypes, developer tools / plugins, and technical learning content. My background includes autonomous underwater-robot development, Sliding Mode Control for robot trajectory tracking, system modeling and simulation, and multi-language control / robotics courseware. Seeking **remote-only** Full-time, Part-time, Contract, Consulting, Research, simulation, software / app / plugin development, and technical-content roles in Control / Robotics.

WORK EXPERIENCE

PERIOD	July 2013 — Present	
EMPLOYER	Self-Employed	Tehran, IRAN
JOB TITLE	Programmer, Engineering Software Developer, Researcher, Technical Educator	
	Delivered 100+ projects for companies and individuals across software / web development, engineering computation, Data Science, Machine Learning, and DevOps. I design engineering-oriented applications, simulations, automation, APIs, desktop / web interfaces, computational tools, and research prototypes; create technical lectures, tutorials, documentation, and educational software / content; and handle requirements, implementation, testing, delivery, and maintenance through Git-based remote workflows.	
PERIOD	January 2011 — August 2011 (Full Time)	
EMPLOYER	Worcester Polytechnic Institute	Worcester, MA, USA
JOB TITLE	Research Assistant	
	Conducted graduate-level Mechanical Engineering research and participated in engineering project work during PhD studies, strengthening research, modeling, technical communication, and interdisciplinary engineering experience. I also hold nine certifications in industrial automation from FESTO.	
PERIOD	July 2007 — August 2010 (Part Time)	
EMPLOYER	Sharif University Of Technology	Tehran, IRAN
JOB TITLE	Project Manager	
	Led a team of 12 graduate / undergraduate students and university staff in development of a laboratory autonomous underwater robot. I coordinated project activities and contributed to software for robot control and monitoring, connecting mechanical-system development with control-oriented software implementation.	

EDUCATION

PERIOD	January 2011 — August 2011		
DEGREE	PhD Candidate in Mechanical Engineering (not finished)		
UNIVERSITY	WPI	Worcester, MA, USA	
PERIOD	October 2007 — September 2010		
DEGREE	Master of Science in Mechanical Engineering		
UNIVERSITY	IRAN University of Science and Technology	Tehran, IRAN	
PERIOD	October 2002 — September 2007		
DEGREE	Bachelor of Science in Mechanical Engineering		
UNIVERSITY	Sharif University of Technology	Tehran, IRAN	

SKILLS

Control / Robotics	Control systems, classical / linear and modern control, system dynamics and modeling, robot kinematics and dynamics, robot control, trajectory tracking, Sliding Mode Control, autonomous / mobile robotics, numerical simulation
Scientific Computing	MATLAB, Simulink, Mathematica, Python, NumPy, SciPy, Matplotlib, Pandas, differential-equation modeling and numerical methods
Programming Languages	C, C++, C#, Python, Java, Go, Rust, Julia, Scala, Ruby, Perl
Engineering UI / Apps	Qt / C++, PyQt, PySide, wxPython, WPF, XAML, JavaFX, desktop engineering tools and visualization interfaces
AI / Perception	Machine Learning, Neural Networks, Deep Learning, Computer Vision, OpenCV, Scikit-Learn, TensorFlow, PyTorch, NLTK, BERTopic
Web / APIs	HTML, CSS, JavaScript, React, Next.js, Three.js, Django, Flask, Node.js, ASP.Net, PHP, REST / API development
Data / Databases	R, SQL, MySQL, SQLite, PostgreSQL, Microsoft SQL Server, MongoDB, data analysis and visualization tooling
Dev Tools	Git, Docker, Linux / Bash, PowerShell, Selenium, Robot Framework, Jenkins, Kubernetes, LaTeX
Project Delivery	PMBOK, Agile, Jira, Trello, Microsoft Project, requirements, documentation, teaching, technical content development

SELECTED CONTROL / ROBOTICS WORK

Control / Robotics Lab / CaaT Academy — Current. Develop structured engineering material and executable examples covering robotics fundamentals, Linear and Modern Control, System Dynamics, robot kinematics / dynamics and control, Advanced Robotics, and Autonomous Mobile Robots using Python, C++, Java, MATLAB, Mathematica, and browser technologies.

Numerical Simulation / Nonlinear Control — Current. Built comparative simulations including an inverted-pendulum cart-pole stabilized with Sliding Mode Control, cross-language numerical integration workflows, differential-equation solver benchmarking, scientific visualization, data pipelines, and simulation-oriented software design. Related work also includes a MATLAB differential-equation integrator benchmark framework.

Autonomous Underwater Robot (AUV). Led a 12-person Sharif University team developing a laboratory AUV and contributed to its control and monitoring software.

Four-Bar Mechanism Simulator. Mechanism-simulation work combining kinematic modeling, geometric constraint solving, visualization, and browser-based engineering interaction.

PUBLICATIONS

Mohammadijoo, A. "Trajectory Tracking of a 2-Link Mobile Manipulator Using Sliding Mode Control Method". *World Academy of Science, Engineering and Technology, International Journal of Mechanical and Mechatronics Engineering*. Vol:17, No:5, 2023

HONORS AND AWARDS

Best Paper Award for paper "Trajectory Tracking of a 2-Link Mobile Manipulator Using Sliding Mode Control Method" at *Robotics and Autonomy Conference*, May 2023 in Amsterdam, Netherlands.

Award-Winning Member for Activities in Marine Engineering Industry at *The 10th National Conference of Iranian Maritime Industries*, 28-30 October 2008 in Abadan / KhoramShahr, IRAN.

WEBSITES

Portfolio	https://mohammadijoo.github.io
GitHub	https://github.com/mohammadijoo
CaaT Academy	https://mohammadijoo.github.io/CaaT_Academy
DevOps Academy	https://mohammadijoo.github.io/DevOps_Academy
Big Data Academy	https://mohammadijoo.github.io/Big_Data_Academy
Youtube	https://www.youtube.com/@abolfazlmohammadijoo
Google Scholar	https://scholar.google.com/citations?user=1re83n8AAAAJ&hl=en
ResearchGate	https://www.researchgate.net/profile/Abolfazl-Mohammadijoo
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